ORIGINAL CONTRIBUTION

Low Rates of Colorectal Cancer Screening in First-Degree Relatives of Our Patients: Are We Failing Them?

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See editorial on page 865.

BACKGROUND: Guidelines recommend screening those with a family history of early-onset colorectal cancer at age 40 years or 10 years before the age of their relative's diagnosis. Currently, there is no literature reporting the screening rate in these individuals, and no protocols are in place to identify and target this population for screening awareness.

OBJECTIVE: This study aimed to assess adherence to current screening guidelines among first-degree relatives of patients with early-onset colorectal cancer.

DESIGN: Retrospective and qualitative study involving a telephone survey where patients were asked about relative's screening status and barriers to screening.

SETTINGS: Two community-based institutions between January 2018 and December 2021.

PATIENTS: Individuals diagnosed with early-onset colorectal cancer who had undergone surgery at our institutions.

MAIN OUTCOME MEASURES: Rate of screening in first-degree relatives of our patients with early-onset colorectal cancer. Other factors measured included demographics, clinicopathologic characteristics, and screening barriers.

RESULTS: Thirty-six patients were identified. The survey response rate was 66.6% (n = 24). A total of 88 first-degree relatives who met the screening criteria resulted in 67.1% of patients (n = 59) having a known screening status. Of the 59 patients with known screening status, only 44% (n = 26) had undergone screening. Patients of Black race, having stage III/IV disease, having Medicare/Medicaid insurance, and living within Baltimore City County were more likely to have family members with unknown or no screening. Lack of insurance coverage was the most common barrier, which was noted in 12.5% of patients (n = 3), whereas 54.1% of patients (n = 13) reported no barriers to screening.

LIMITATIONS: Retrospective design.

CONCLUSIONS: Most first-degree relatives of patients diagnosed with early-onset colorectal cancer do not undergo colorectal cancer screening. This could be attributed to the lack of protocols that could guarantee these individuals are informed of their elevated risk and the different options available for screening. Furthermore, our study suggests that racial and socioeconomic disparities exist among high-risk patients who should pursue screening. See **Video Abstract**.

BAJAS TASAS DE DETECCIÓN DEL CÁNCER COLORRECTAL EN LOS FAMILIARES DE PRIMER GRADO DE NUESTROS PACIENTES: ¿LES ESTAMOS FALLANDO?

ANTECEDENTES: Las directrices recomiendan realizar pruebas de detección a las personas con antecedentes familiares de cáncer colorrectal de aparición temprana a los 40 años o 10 años antes de la edad del diagnóstico de su familiar. Actualmente, no hay literatura que informe la tasa de detección en estos individuos y no existen

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protocolos para identificar y dirigirse a esta población para concientizar sobre la detección.

OBJETIVO: Evaluar el cumplimiento de las pautas de detección actuales entre los FDR de pacientes con cáncer colorrectal de aparición temprana.

DISEÑO: Estudio retrospectivo y cualitativo que incluyó una encuesta telefónica en la que se preguntó a los pacientes sobre el estado de detección de sus familiares y las barreras para la detección.

AJUSTES: Dos instituciones comunitarias entre enero de 2018 y diciembre de 2021.

PACIENTES: Personas diagnosticadas con cáncer colorrectal de inicio temprano que habían sido intervenidas quirúrgicamente en nuestras instituciones.

PRINCIPALES MEDIDAS DE RESULTADO: Tasa de detección en familiares de primer grado de nuestros pacientes con cáncer colorrectal de aparición temprana. Otros factores medidos incluyeron datos demográficos, características clínico-patológicas y barreras de detección.

RESULTADOS: Se identificaron treinta y seis pacientes. La tasa de respuesta a la encuesta fue del 66,6% (n = 24). Resultaron un total de 88 familiares de primer grado que cumplieron con los criterios para la detección, y el 67,1% (n = 59) tenía un estado de detección conocido. De los 59 con estado de detección conocido, se informó que solo el 44% (n = 26) se había sometido a pruebas de detección. Los pacientes de raza afroamericana, enfermedad en etapa III/IV, Medicare/Medicaid y que vivían dentro del condado de la ciudad de Baltimore tenían más probabilidades de tener familiares con pruebas de detección desconocidas o sin ellas. La falta de cobertura de seguro fue la barrera más común observada por el 12,5% (n = 3); mientras que el 54,1% (n = 13) no informó ninguna barrera para el cribado.

LIMITACIONES: Diseño retrospectivo.

CONCLUSIONES: La mayoría de los familiares de primer grado de pacientes diagnosticados con cáncer colorrectal de aparición temprana no se someten a pruebas de detección de cáncer colorrectal. Esto podría atribuirse a la falta de protocolos que garanticen que estas personas estén informadas sobre su elevado riesgo y las diferentes opciones disponibles para el cribado. Además, nuestro estudio sugiere que existen disparidades raciales y socioeconómicas entre los pacientes de alto riesgo que deberían someterse a pruebas de detección. *(Traducción—Dr. Francisco M. Abarca-Rendon)*

KEY WORDS: Barriers; Early-onset colorectal cancer; First-degree relatives; Screening; Survey.

Colorectal cancer (CRC) is a significant contributor to morbidity and mortality in the United States, ranking third in both cancer incidence and cancer deaths.[1] Screening tools such as colonoscopies have been estimated to reduce incidence by 40% and mortality by 60%; therefore, early detection is critical to improving patient outcomes.[1] Interestingly, although there has been a recent decrease in overall CRC incidence, there has been a marked increase in early-onset colorectal cancer (EOCRC), defined as CRC diagnosed before the age of 50 years.[2]

Family history is considered an important risk factor. Having a first-degree relative (FDR) with EOCRC has been associated with a 2-fold increased risk of acquiring the disease.[3] In addition, individuals with a positive family history tend to develop CRC at a much earlier age.[4] Given these findings, major society guidelines have recommended that individuals with 1 or more affected FDRs start screening for CRC at age 40 or 10 years before the age of their relative's diagnosis.[5] Although the National Comprehensive Cancer Network guidelines advise assessing susceptibility risk in all patients with CRC and their family members, it remains unclear to what extent these guidelines are being implemented in clinical practice.

Some studies assessed the screening rates among FDRs of patients diagnosed with CRC. Of the literature published, it is estimated that fewer than half of individuals with an affected FDR adhere to screening guidelines.[3,6–8] A study by Bujanda et al[9] identified an adherence rate to colonoscopy screening of only 38% among FDRs of patients with CRC in Spain. Interestingly, adherence was noted to be higher in relatives younger than 65 years, female relatives, and siblings or children.[9] Another study based on the Australian CRC screening guidelines reported an adherence rate of 69% among FDRs who had received some form of CRC testing. Factors such as sex, education level, marital status, and living in Metropolitan areas influenced adherence to screening guidelines.[10] In addition, a systematic review aimed at identifying barriers to CRC screening among FDRs showed that fear of cancer diagnosis, sociocultural factors, negative perceptions of screening tests (embarrassment, cost, and accessibility), and lack of awareness about increased risk were the most reported factors.[11]

Although the aforementioned studies focused on FDR of patients with CRC, the distinction of EOCRC was not highlighted. This is of particular concern, as targeting this high-risk population is critical to reducing the morbidity and mortality associated with this disease. Thus, the objectives of this study are to assess the adherence of FDRs of patients with EOCRC to current CRC screening guidelines, as well as to identify specific barriers to screening uptake in this population. The results of this study can help inform the development of more robust CRC screening strategies that better target this high-risk population.

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MATERIALS AND METHODS

Study Design

After approval of the institutional review board, our enterprise-wide medical record system was queried for all adult patients (older than 18 years and younger than 50 years) who were diagnosed with CRC and had undergone surgery at either of our 2 index institutions, MedStar Franklin Square Medical Center and MedStar Good Samaritan Hospital, between January 2018 and December 2021. These hospitals have an average of 90 and 40 CRC cases per year, respectively.

After informed consent was obtained, we conducted a telephone survey (Fig. 1) between January and March 2022 where, with the patient's help, we identified their FDRs who met CRC screening criteria and worked to obtain completed questionnaires focusing on these relative's screening status and barriers to screening. This survey was elaborated by the members of the study team, and each question was formulated to facilitate the obtaining of pertinent information required to reach the study's objectives. Each patient completed the survey 2 times, with a 1-month interval between both interviews and with a different member of our team. Both members conducting the survey were part of the institution where the study was performed, yet none had directly been in contact with the study participants during their time of treatment. Data gathered were de-identified and stored in password-protected institutional software.

The primary end point was the rate of screening in the FDRs of our patients with EOCRC, with screening methods including any of the tests currently recommended by the American College of Gastroenterology (fecal immunohistochemical test, flexible sigmoidoscopy, stool DNA test, colonoscopy, and CT colonography). Secondary end points included barriers to screening, demographics of patients with EOCRC, rate of screening by race, and socioeconomic status.

The decision to focus on the FDRs instead of on our patients themselves was made on the basis of accessibility and institutional policies. The study's FDRs are not our institution's patients; therefore, our institutional review board does not have jurisdiction over their information.

Although major guidelines recommend colonoscopy as the method of choice for screening in high-risk individuals, we opted to include all screening methods because this will give a better overall estimate of how many of these relatives are undergoing some form of screening.

Statistical Analysis

Baseline demographic and socioeconomic patient characteristics were collected from patients' electronic medical records. Variables included age, sex, race, insurance status, and clinicopathologic characteristics of their cancer. Categorical variables were reported as frequency (%), and qualitative variables were reported as mean and median with interquartile range.

RESULTS

Overall Screening Rate

Thirty-six patients who met inclusion criteria were identified (Fig. 2). The survey response rate was 66.6% (n = 24). A total of 88 FDRs were identified, with 67.1% (n = 59) of them having a known screening status. Among 24 respondents, 13 family members were reported as

Assessment of CRC screening in high-risk family members of patients with EOCRC (survey)

1. At what age were you diagnosed with EOCRC?
2. How many first-degree relatives (parent, sibling, or child) older than 40 y, or up to 10 y younger than the age at which you were diagnosed with CRC, do you have?
3. Which of these family members have had CRC screening?
4. What method was used to perform this screening?
5. Are there any barriers to having this screening performed?
6. Was there an attempt to reach out and inform relatives of the risk of colon cancer?
7. If yes, how did they learn about their high-risk status and need for screening?
    a. PCP
    b. Your GI or surgeon
    c. Their GI
    d. Advertisements/campaigns
8. Do you believe members of your family would be open for a discussion on the risk of colon cancer and the need for screening?
9. What methods of screening do you know of?

FIGURE 1. Survey questionnaire. CRC = colorectal cancer; EOCRC = early-onset colorectal cancer; PCP = primary care provider.

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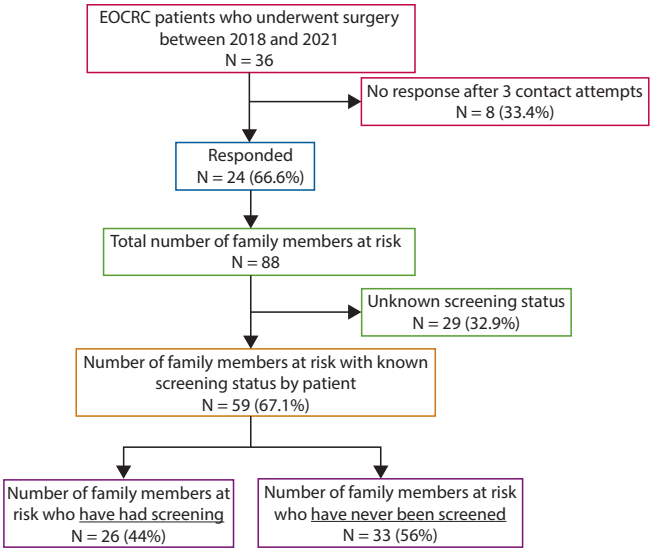


FIGURE 2. Overall survey response rate and rate of colorectal cancer screening in first-degree relatives of patients with EOCRC. EOCRC = early-onset colorectal cancer.

deceased at the time of the survey: 8 parents (5 died at an age older than 60 years due to medical conditions, 2 had accidental deaths at a young age, and 1 died at age 54 years due to pancreatic cancer), 4 siblings (2 accidental deaths at a young age, 1 at age 42 years from breast cancer, and 1 at age 50 years from stroke complications), and 1 child with an accidental death at age 22 years. Of the 59 FDRs with known screening status, it was reported that only 44% (n = 26) had undergone screening. Colonoscopy was the only screening method used by any of the FDRs.

Demographics and Clinicopathologic Characteristics of Patients With EOCRC

Out of our total cohort of patients, 50% were men (n = 18), 69.4% (n = 25) were White (versus 25% [n = 9] Black), and all had English as their primary language. The median age at diagnosis of CRC was 43 years, and 13.9% (n = 5) of patients had a family history of CRC. Patient demographics are summarized in Table 1.

Regarding their cancer diagnosis, 69.4% (n = 26) of patients had stage III/IV disease, with the main presenting symptom being hematochezia (n = 15); the primary site had a similar distribution with 52.7% (n = 19) having colon cancer and 47.3% (n = 17) having rectal cancer. Low anterior resection and abdominoperineal resection were the most performed procedures in 27.7% (n = 10) of patients each. The clinicopathologic characteristics of the patients are summarized in Table 2.

Survey Results

Based on the patients who responded (n = 24), the survey results indicated that 54.1% (n = 13) reported no barriers to their family members getting screened, and for 41.6% (n = 10) of those who did report barriers, lack of insurance coverage was the most noted by patients (n = 3), followed by unrealistic fear of the result (n = 2), laziness (n = 2), and lack of knowledge on how to decide/plan screening (n = 2). Eighty-three percent of our patients (n = 20) reported that their FDRs were informed of their high-risk status. Of these, 19 stated that their FDRs were informed by patients themselves, with only 1 patient citing their primary care provider relaying this information. In addition, 100% (n = 24) of our patients reported that there was no attempt by a medical professional from the treating team to reach their family members and make them aware of their high-risk status. Furthermore, 66.6% (n = 16) admitted to only knowing colonoscopy, with only 20.8% (n = 5) knowing both colonoscopy and Cologuard, and 12.5% (n = 3) not knowing of any screening method (Table 3).

When stratified into relatives with positive screening versus relatives with negative/unknown screening, we found that Black patients were more likely to have a FDR with negative/unknown screening compared to their White counterparts (83.3% vs 50%; $p = 0.234$). FDRs of patients with private insurance were more likely to have had screening than those with Medicare/Medicaid

TABLE 1. Demographics of the study patients with early-onset CRC (N = 36)

Variables	*n (%)*
Age at diagnosis, y, median (IQR)	43 (38.3–46)
Sex	
Male	18 (50%)
Race	
White	25 (69.4)
Black	9 (25)
Asian	2 (5.6)
Have health insurance	36 (100)
Insurance type	
Medicare/Medicaid	20 (55.6)
Private	16 (44.4)
Family history of CRC	5 (13.9)
Smoking	
Never	17 (47.2)
Active	12 (33.3)
Former	7 (19.4)
Alcohol	
Occasional	13 (36.1)
Never	23 (63.9)
Primary language	
English	36 (100)
Occupation	
Unemployed	8 (22.2)
Vocational degree	8 (22.2)
Managerial/administrative position	6 (16.7)
Sales position	4 (11.1)
Service industry	2 (5.6)
Technician	3 (8.3)
Other (sailor, dancer)	2 (5.6)
Unknown	3 (8.3)

Data presented as n (%) unless otherwise indicated.
CRC = colorectal cancer; IQR = interquartile range.

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TABLE 2. Clinicopathologic characteristics of the study patients with early-onset colorectal cancer (N = 36)

Variables	*n (%)*
Presenting symptom	
Hematochezia	15 (41.7)
Abdominal pain	5 (13.9)
Bowel obstruction	6 (16.7)
Iron deficiency anemia	2 (5.6)
Incidental	2 (5.6)
Other	6 (16.7)
Stage at diagnosis	
Stage I	3 (8.3)
Stage II	8 (22.2)
Stage III	18 (50)
Stage IV	7 (19.4)
Primary site	
Colon	19 (52.7)
Rectum	17 (47.3)
CEA, ng/mL, median (IQR)	3.3 (1.6–4.2)
Surgery	
Right/extended right hemicolectomy	6 (16.7)
Sigmoidectomy	3 (8.3)
Low anterior resection	10 (27.7)
Abdominoperineal resection	10 (27.7)
Total proctocolectomy	4 (11.1)
Palliative diversion	3 (8.3)
Approach	
Open	9 (25)
Laparoscopic	8 (22.2)
Robotic	19 (52.8)
Genetic testing performed	16 (44.4)
MSI/MMR	
Intact	33 (91.7)
Not performed	2 (5.6)
MLH1$^-$, PMS2$^-$, MSH6$^-$, MSH2$^-$	1 (2.7)

Data presented as n (%) unless otherwise indicated.
IQR = interquartile range; MSI/MMR = microsatellite instability and mismatch repair.

TABLE 3. Survey results including screening barriers, screening methods known by patient, and family member knowledge of high-risk status (N = 24)

Items addressed by survey questions	*n (%)*
Screening barriers	
Yes	10 (41.6)
Insurance	3 (12.5)
Unrealistic fear/prefer not to know	2 (8.3)
Laziness	2 (8.3)
Unsure how to go about it	2 (8.3)
Refusal to be NPO at midnight	1 (4.2)
No	13 (54.1)
Not sure	1 (4.2)
No. of FDRs at risk	
0–1	2 (8.3)
2	5 (20.8)
3	6 (25)
4	6 (25)
≥5	5 (20.8)
Distribution of FDRs at risk	
Parents only	4 (16.6)
Siblings only	7 (29.3)
Multiple family members	13 (54.1)
Provider attempts to inform family members of high-risk status	0 (0)
Family member informed of high-risk status	
No	4
Yes, informed by:	
Patient	19
Primary care physician	1
No. of FDRs undergoing screening	
Does not know	9
0	5
1 of 4	1
1 of 6	1
2 of 3	1
2 of 4	1
2 of 5	1
All	4
Screening methods known by the patient	
Colonoscopy only	16 (66.6)
Colonoscopy and Cologuard	5 (20.8)
Not sure	3 (12.5)

FDR = first-degree relative; NPO = nil by mouth.

insurance (50% vs 33.4%; $p = 0.34$). Similarly, patients with stage I/II cancer were more likely to have a FDR undergoing screening than those with stage III/IV cancer (42.9% vs 28.6%; 0.643; Table 4). When classified by zip code, patients within Baltimore City County had the lowest rate of FDRs undergoing screening compared to other counties in Maryland 6 (22.20%). Baltimore City County had the lowest median family income compared to others in the study. Other counties in decreasing order of patients with percentage of FDRs undergoing screening included Anne Arundel County 4 (100%), Howard County 4 (80%), and Baltimore County 15 (33%).

DISCUSSION

As EOCRC has seen a marked increase in incidence in recent years, early detection has become of the highest priority.[12] This is true for both the general population, evident by the recent update on recommended screening age from 50 to 45 years, and for the ever-growing body of high-risk patients based on family history that results from this incidence increase. Our study showed that fewer than half of the first-degree family members of our patients with EOCRC who meet appropriate criteria have undergone screening.

Within our patient cohort, we noted that almost half of the patients had rectal cancer. Although this is different from overall colon and rectal cancer incidence ratios in the general population, we hypothesize that this difference arises from the population that was the focus of our study because multiple studies show that the higher incidence of colon and rectal cancer in patients with EOCRC is mainly driven by rectal cancer cases.[13–15]

Several screening barriers contributing to low compliance have been reported in the literature. A 2019 survey on patients' self-reported barriers showed lack of

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TABLE 4. Patient variables stratified into undergoing screening vs unknown/no screening

Variables	*Unknown/no screening (N = 14)*	*Undergoing screening (N = 10)*	*p*
Age at diagnosis, y, median (IQR)	43.5 (37.7–48)	43.4 (38.7–46.5)	0.99
Sex			
Male	6 (42.9)	4 (40)	0.611
Race			
White	8 (57.1)	8 (80)	0.234
Insurance type			0.34
Medicare/Medicaid	8 (57.1)	4 (40)	
Private	6 (42.9)	6 (60)	
Family history of CRC			
Smoking	7 (50)	5 (50)	0.66
Alcohol	5 (35.7)	4 (40)	0.582
Occupation			
Vocational/managerial degree	3 (23.1)	5 (50)	0.184
Stage at diagnosis			
Stage III/IV	10 (71.4)	7 (70)	0.643

Data presented as n (%) unless otherwise indicated.
CRC = colorectal cancer; IQR = interquartile range.

insurance/coverage and fear of the examination as main reasons for noncompliance with CRC screening.[16] This was consistent with our findings on barriers for FDRs. Most importantly, 2 of our patients cited a lack of knowledge on how to decide/plan screening as being an obstacle for their family members. This is further highlighted by the fact that most of our patients recalled that no medical professional from the treating team reached out to their family members to inform them of their high-risk status based on family history, with most patients reporting that they carried out this task by themselves. We hypothesize that the 26 patients who underwent screening did so because of multiple factors, including 1) our patients informing them of their high-risk status, 2) the awareness of the FDRs regarding our patient's cancer diagnosis and treatment, prompting self-care, and 3) the FDRs' own knowledge of current guidelines and recommendations. The main point made by our study is that, based on our results, most of the FDRs of our patients rely on the latter's knowledge and ability to pass down information to take ownership of their own health care. When considering that an overwhelming majority of our patients showed limited knowledge of the different screening methods, we argue that lack of appropriate education/guidance is potentially the biggest barrier to screening for these FDRs.

Racial and socioeconomic disparities in screening also have been well documented. Historically, individuals of Black race and low socioeconomic status have been shown to have lower screening rates than individuals of White race and high socioeconomic status.[17] When it comes to FDRs in this study, our analysis showed similar results. Recently, Shah et al[18] showed that some of these demographic disparities might be changing when his study, using a national database, reported higher rates of screening among Black individuals. The difference in our results might be attributed to our study design, which focused on a specific geographic area. This finding highlights that screening needs to be individualized to the population being served.

There are multiple strategies that help improve screening adherence for the general population. In 2020, Dougherty et al[19] published a systematic review and meta-analysis evaluating interventions intended to increase CRC screening in the United States. A total of 73 studies were reviewed, and interventions were classified as patient-directed and clinician-directed. Results of the review showed that interventions involving patient navigation (eg, guiding the patient through health care system barriers) and fecal blood test outreach (eg, active fecal blood test distribution) were the most effective at improving screening rates.[19] Other studies have shown good results after interventions focused on both provider education[20] and patient education.[21,22]

Overall, having an efficient system that combines these methods with the addition of appropriate follow-up (ie, patient reminders) could potentially result in great increases in CRC screening.[23] In 2018, Hassan et al[24] published a study in which they advocated that several, usually multifaceted, interventions must be undertaken to promote and facilitate CRC screening. Within their study, they recommended a cycle of different steps to take to ensure appropriate screening (engage, reinforce, target, tailor, facilitate, and motivate).[24]

As a group that is at particularly high risk, the FDRs of our patients with EOCRC represent a vulnerable population, which should be the focus of our screening efforts. Although multiple studies have been conducted to evaluate the effectiveness of different interventions in improving screening for the general population, no standardized protocols have been formally implemented. This is possibly associated with increased costs and a lack of human resources to maintain such programs. Focusing on a limited group of patients, like the ones in our study, could prove

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successful at officially implementing and maintaining these interventions throughout various clinical institutions.

The current guidelines from the National Comprehensive Cancer Network advise assessing the genetic susceptibility risk in all patients with CRC. With the advancement of multigene panel testing and whole exome sequencing, a direction toward genome profiling beyond universal testing for mismatch repair (MMR) deficiency, especially among younger patients, has been advocated in the literature.[25,26] A study by You et al[27] that aimed to implement a universal germline cancer risk assessment and testing strategy among patients with early-onset CRC showed that 1 in 5 patients had inheritable pathogenic mutations linked to cancer predisposition, often affecting DNA recombination or damage repair. Even patients without apparent family history or MMR deficiency still had a significant rate of these mutations.[27] Pearlman et al[28] conducted a landmark study as part of the Ohio Colorectal Cancer Prevention Initiative focusing on implementing universal genetic testing for patients younger than 50 years diagnosed with CRC to identify hereditary cancer syndromes and enhance CRC screening compliance. Their findings revealed that 16% of patients carried mutations associated with cancer predisposition. These mutations included Lynch syndrome, familial adenomatous polyposis, and others not previously linked to CRC risk.[28] In 2021, the same group showed that 7.1% of patients with CRC had pathogenic germline variants, including those in MMR and non-MMR genes that would have been missed by universal tumor screening for MMR deficiency alone, suggesting that pan-cancer multigene panel testing should be considered for all CRCs.[29]

Finally, our findings should be considered in the context of the following limitations. First, screening history for the FDRs was obtained via our patients, which may give way to recall bias and misinformation. Second, our small sample size limited our ability to determine the statistical significance of our results. Third, as evident by the number of cases per year at our index hospitals, this study was conducted at relatively low-volume centers, limiting the generalizability of our results. However, we believe that given the paucity of data on this topic, the relevance of this study persists. In addition, we believe that even larger centers with high referrals, and patients who go on to obtain postoperative integrated care elsewhere, might still face issues with continuity of care and outreach to family members a risk. Finally, we must account for the health literacy of our patients and their FDRs as a potential limiting factor because it could influence awareness and subsequent screening behaviors.

CONCLUSION

Most FDRs of patients diagnosed with EOCRC do not undergo CRC screening. This could be attributed to the lack of protocols or guidelines that could guarantee that individuals are informed of their elevated risk and the different options available for screening. Furthermore, our study suggests that racial and socioeconomic disparities exist among these patients who should seek screening. In addition, this study highlights the importance of proper education when it comes to adherence to screening guidelines.

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